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MILK Symposium review: Community-tailored training to improve the knowledge, attitudes, and practices of women regarding hygienic milk production and handling in Borana pastoral area of southern Ethiopia*

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ABSTRACT

Milk and milk products are essential in the diets of the Borana pastoral community in Ethiopia. Traditional handling and processing of dairy products using basic equipment and infrastructure coupled with a preference for raw milk consumption pose potential health risks to consumers. We tested the effect of an intervention designed to improve the hygienic handling and safe consumption of milk on the knowledge, attitudes, and practices (KAP) of women who produce and sell dairy products. The intervention consisted of 16 h of training on good milk production practices and prevention of milk-borne diseases. A total of 120 women were trained and their KAP assessed at baseline (pretraining), immediately after training, and 6 mo after training. Overall, training increased the knowledge score of the participants from 75.6 to 91.4% in the immediate post-training assessment, and to 90.0% at 6 mo post-training. Compared with pretraining (58.8%), we found a statistically significant difference in the overall attitude score at the immediate post-training evaluation (64.7%) but not 6 mo after (61.4%). We observed a similar increase in the understanding of correct practices from 49.5% at pretraining to 64.7% 6 mo following the training. For some desirable attitudes and practices, the proportion of women reporting adoption at pretraining was low and the change derived from training still left one-third of respondents displaying a negative attitude and a quarter of them reporting wrong practices. We recommend that future training interventions be complemented with locally adaptable technologies, provision of incentives, and creation of an enabling environment including improved access to clean water and sanitation facilities to affect not only knowledge, but also attitudes and ultimately practices in the long term.

Key words: milk hygiene, community training, pastoral community, Borana

INTRODUCTION

In most of the developing countries of the world, like Ethiopia, important technical, economic, and cultural constraints create suboptimal conditions for the hygienic production of milk (Schoder et al., 2013; Kamana et al., 2014; Ledo et al., 2019). As a result, milk and milk products can harbor a variety of food-borne pathogens (Oliver et al., 2009; Dhanashekar et al., 2012) that can cause diseases in humans if the products are consumed without boiling or pasteurization (De Buyser et al., 2001; Bankole et al., 2011). Milk-borne pathogens cause human diseases ranging from gastrointestinal disturbances characterized by diarrhea and vomiting (Hetzel et al., 2004) to other generalized and even life-threatening food-borne illnesses such as hemolytic uremic syndrome associated with *Escherichia coli* O157:H7 infection (Lye et al., 2013).

Various factors ranging from premilking conditions (e.g., the health status of the lactating animals) to postmilking conditions (e.g., improper handling, storage, and processing) are responsible for low microbiological quality and safety of milk and milk products (Millogo

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et al., 2010; Te Giffel and Wells-Bennik, 2010; de la Fuente and Juárez, 2015). One approach to improve the quality and safety of milk and milk products is educating the people involved along the milk production and supply chains on proper hygiene and good production practices. However, studies assessing the effectiveness of training on changing the knowledge, attitude, or behavior of the people toward improving the quality and safety of milk and milk products are limited in most developing countries (Alonso et al., 2018; Lindahl et al., 2018a), especially in the most challenging contexts such as production in remote rural areas. In the case of African countries, it is particularly in such contexts that women are responsible for the daily routines of traditional dairy farming. However, these women often do not get the opportunity for training and capacity building (Tangka et al., 2000; Kristjanson et al., 2010; Edemo, 2017). Training on dairy hygiene and good production practices can have a positive effect by increasing the knowledge of actors (Lindahl et al., 2018b); therefore, when women are the main operators in milk production, to be successful and sustainable, training initiatives should target women.

Borana pastoral area is situated in the southern part of Ethiopia and is characterized by arid and semi-arid climates, with livestock production being the mainstay of the people. Milk is a common component of the diet in pastoral households and an important source of nutrients for the family (Megersa et al., 2014). Milk consumption behavior of the pastoral community is characterized by a preference for fresh raw milk and spontaneously fermented raw milk products which, given the bacterial load of the final product, could pose a risk to the health of consumers (Amenu et al., 2019b). The preference for raw milk over boiled, packaged, pasteurized, or refrigerated milk (when these are available) is one of many potential risks that could affect the health of milk consumers. A high prevalence of mastitis in livestock and ignorance or misconceptions about mastitis and occurrence of zoonotic diseases such as bovine tuberculosis and brucellosis in the area are well-known potential risk factors for milk-borne disease in humans. Previous studies conducted in Borana have reported a high prevalence of mastitis in various livestock species commonly kept in the area: 15.5% in goats (Megersa et al., 2010), 59.1% in cattle (Adane et al., 2012), and 44.8% in camels (Regassa et al., 2013). There is a widespread misperception among the Borana pastoral community regarding the causes of acute mastitis. For example, the community largely perceives that mastitis in livestock (especially in cows) is caused by the "evil eye" affliction, and the community often resorts to "magical" options for treatment. This perception makes it very difficult to control animal mastitis in the community, which in turn can lead to poor animal welfare and low quality and safety of milk as a result of mastitis-causing pathogens secreted in the milk (Amenu et al., 2017).

In addition to mastitis, milk-borne zoonoses such as brucellosis and tuberculosis are prevalent in Borana pastoral area (Megersa et al., 2012; Duguma et al., 2017). Megersa et al. (2012) reported an 8% seroprevalence of brucellosis in cattle, 1.8% in camels, and 1.6% in goats. In a recent epidemiological investigation using a comparative intradermal tuberculin test, bovine tuberculosis was reported in 3.8% of cattle in Borana (Duguma et al., 2017). Moreover, microbiological assessment of 203 samples of milk and fermented milk products in the area detected *E. coli* O157:H7 and *Staphylococcus aureus* in 2.5 and 10.8% of the samples, respectively (Amenu et al., 2019a). In general, risky milk consumption behavior of the pastoralists, such as high preference for raw milk, coupled with the occurrence of zoonotic diseases, can present a significant health burden to the community (Amenu et al., 2019b). The objective of this study was to evaluate the effect of community-tailored training on the knowledge, attitude, and practices (**KAP**) of women in the Borana pastoral area of southern Ethiopia with regards to milk production, handling, and consumption. We hypothesized that training will improve the KAP of the women with regard to milk handling and consumption as well as prevention of milk-borne zoonotic diseases.

MATERIALS AND METHODS

Study Area

The study was carried out in the Borana pastoral area of southern Ethiopia. This area is known for livestock production, which plays economic, social, and cultural roles in the community. Traditionally, cattle have been the preferred livestock species reared for milk production in Borana. However, rearing of dairy camels and goats has become increasingly popular among pastoral communities in response to major drought events experienced over the past decades among the pastoral communities, as this gives them more resilience. Milk from cattle is consumed raw or surplus milk is spontaneously fermented into traditional dairy products. Fermented whole milk (*itittuu*) is the preferred processed milk product in the area. Camel milk is consumed largely in raw form and commonly sold in the local market or taken to Moyale, a town on the border with Kenya, for sale. Goat milk is consumed by mixing it with boiled tea. Detailed descriptions of the study area, as well as milk production and consumption practices, are reported elsewhere (Amenu et al.,

2019b). Study protocols were reviewed and approved by the Institutional Research Ethics Committee of the International Livestock Research Institute (Nairobi, Kenya; reference number ILRI-IREC2016–20) and by the Oromia Regional Health Bureau (reference number BEFO/AHDh/1-67/246).

Identification and Selection of Women Trainees

Pastoral women selling traditionally fermented yogurt were identified at 3 market places (Yabello, Dubuluk, and Elweya) in the study area. Women were targeted in this study because they are customarily involved in milking and handling of milk and dairy products. All women selling milk or dairy products in the markets on the days of our visits and having dairy animals (camels or cattle owned by them or their households) were eligible and invited to participate in the study. After we explained the objectives of the training and obtained their verbal consent, we recruited women into the study. The identified women were asked to provide the names of other women who handle milk products in the markets but were not present on the day of our visit. These additional women were contacted and consent was obtained in a similar fashion. Once specific dates and venues were set for the training, the participants were informed accordingly. A total of 120 women from Yabello (n = 60), Elweya (n = 30), and Dubuluk (n = 30) markets were trained. The sample size was calculated to detect a 30% increase in the mean KAP scores of the training group (to reach 70%) after the training, assuming 40% at baseline with a study power of 87%, standard deviation of 1.4, and 5% statistical significance.

Training Guide and Content Preparation

The training guide was prepared by consulting various food and dairy hygiene training manuals and textbooks (Lore et al., 2006; WHO, 2006; Farmer's Weekly, 2012; Goopy and Gakige, 2016) and aligning the content to suit the context of the trainees, their educational backgrounds, and livestock production practices in the area. For example, given that most women did not have a formal education, the use of technical terms was minimized. Furthermore, the content was focused on practices that can be implemented in the local context, considering the pastoralists' lack of access to advanced dairy technologies. The content of the training was further refined based on previous qualitative studies that assessed the socio-cultural aspects of udder health management in livestock and milk handling and milk consumption practices in the study area (Amenu et al., 2017, 2019b). The final training guide included 5 modules: (1) general food hygiene and food-related health problems, (2) the importance of milk in human nutrition and factors affecting the contamination of milk and milk products, (3) udder health problems and their management, (4) how to produce clean and safe milk, and (5) prevention of milk-borne zoonotic diseases, specifically brucellosis and tuberculosis. The detailed contents of the training guide are available elsewhere (Amenu et al., 2018).

Approach and Implementation of the Training Intervention

The training was delivered in collaboration with the Yabello Pastoral and Dryland Agriculture Research Centre and the Pastoral Development Offices of Yabello, Elweya, and Dubuluk districts from September 19 to October 10, 2018. Four trainings were organized, each having 30 participants. Each training was delivered over 3 consecutive days, in which 2 half days were devoted to pre- and post-training evaluation and 2 full days (16 h) to content delivery. The training approach consisted of interactive lectures followed by questions and answers with intensive discussions. The materials used in the training included PowerPoint presentations, posters (for tuberculosis and brucellosis), flash cards with pictures, and short videos showing handling practices for milk in the area, possible sources of contamination, and the means to avoid them. To maintain consistency of the training, all lectures were delivered by one person fluent in the local language (Afaan Oromoo) with close supervision from the main trainer (the first author of this paper). The person who delivered the lectures is a native Borana veterinarian working with the Elweya Pastoral Development Office, with over 6 yr of work experience in the area. Pretraining preparations (i.e., the content to be covered every day was revised jointly by the main trainer and the training facilitator, and the methods to be used for each training component agreed upon and practiced before the trainings) were done to ensure consistency in the delivery of the content between the various trainings. The training was offered free to participants, with the research project also covering expenses associated with transportation of and refreshments for participants.

Pre- and Post-Training Assessments

Knowledge, attitude, and practices were assessed using a questionnaire specifically designed for this purpose, taking into account the content of the training guide. The questionnaire was administered individually as an interview that took 15 to 20 min to complete. Two post-training assessments were conducted, one im-

mediately after the training and the second one 6 mo after training. The questionnaire had 4 sections: (1) general characteristics of the respondents; (2) 8 knowledge questions (with answer options of "true," "false" and "I don't know"; Table 1); (3) 10 attitude questions (3 possible responses: agree, neutral, and disagree; Table 2); and (4) 9 practice questions (with the responses indicated as always, sometimes, or never performing a given practice; Table 3). The knowledge questions were presented either as true or false statements in which a "yes" answer for true statements and a "no" answer for false statements were regarded as the "correct" answer. Attitude questions were worded either positively or negatively. Having a "positive attitude" was defined as either agreeing with a positively worded statement or disagreeing with a negatively worded statement. A "negative attitude" was defined as either disagreeing with a positively worded statement or agreeing with a negatively worded statement. "Neutral" responses were categorized as negative attitude.

Data Management and Statistical Analysis

The data from the KAP pre- and post-training assessments were recorded in Microsoft Excel 2007 (Microsoft Corp., Redmond, WA) and later imported into Stata 15 (Stata Corporation, College Station, TX) for statistical analysis. The demographic and livestock keeping characteristics of the study participants were summarized using means and standard deviations for continuous variables and percentages for categorical variables. Answers to knowledge questions were coded as 1 if correct and 0 if wrong or if the response was "I don't know." Similarly, "positive attitudes" (see previous section) were coded as 1 and "negative attitudes" were coded as 0. For practice questions, practices that the respondents said they performed "always" were coded as 1, whereas those they said they performed "sometimes" or "never" were coded as 0. This approach in which semi-correct, incorrect, or neutral answers for KAP were classified as incorrect was based on the methodology used in other studies (Choudhury et al., 2011). The effect of the training was evaluated by comparing the baseline knowledge and attitude scores (pretraining) with the respective post-training scores (on the last day of the training and after 6 mo). In the case of practices, the comparison was made between pretraining and 6 mo post-training assessment. Two main comparisons were made: per statement and overall scores of KAP. For each statement on KAP, the proportion of women who answered correctly was calculated. The McNemar Chi-squared test (paired test for before/after comparisons) was used to compare responses between the post-training assessments and the pretraining as-

Table 1. Knowledge of pastoral women about milk hygiene and zoonoses prevention: comparing post-training evaluations (immediately and 6 mo after training) with pretraining (baseline) status[1]

		Immediate post- vs. pretraining comparison						6 mo post- vs. pretraining comparison				
		Pretraining		Immediate post-training				6 mo post-training				
Code	Knowledge questions	n	%	n	%	McNemar χ^2	*P*-value	n	%	McNemar χ^2	*P*-value
K1	People can get a disease through consumption of food	118	86.4	118	99.2	15.0	<0.001	118	99.2	15.0	<0.0001
K2	Raw food is more likely to cause illness than cooked food	118	94.1	118	100	7.0	0.008	118	99.2	4.5	0.034
K3	Eating food contaminated with feces from animals can make people sick	118	95.8	118	99.2	2.7	0.10	118	100	5.0	0.025
K4	Most of the illness we get from food is caused by germs	118	84.7	118	100	18.0	<0.001	118	100	18.0	<0.0001
K5	All people, including children, adults, pregnant women, and elders are at equal risk of food-borne disease[2]	118	39.0	117	38.5	0.1	0.83	117	38.5	0.0	1.00
K6	Having dirty hands when milking animals compromises quality and safety of the milk	117	94.0	118	100	7.0	0.008	118	100	7.0	0.008
K7	Tuberculosis can be transmitted from animals to humans	118	54.2	117	97.4	47.3	<0.001	118	89.8	33.9	<0.0001
K8	People can get serious disease when touching aborted materials of animals	118	57.6	118	98.3	48.0	<0.001	118	94.1	36.3	<0.0001

[1]Where n = number of respondents with valid responses to each question, and % = proportion of women responding correctly to the statement.
[2]Indicates false statements.

Table 2. Attitudes of pastoral women regarding milk hygiene and zoonotic diseases: comparing post-training evaluations with pretraining evaluation[1]

		Immediate post- vs. pretraining comparison						6 mo post- vs. pretraining comparison			
		Pretraining		Immediate post-training		McNemar		6 mo post-training		McNemar	
Code	Attitude statements	n	%	n	%	χ^2	*P*-value	n	%	χ^2	*P*-value
A1	Consuming raw milk makes you stronger[2]	118	16.9	118	64.4	49.0	<0.001	118	44.9		<0.001
A2	Children should not be given raw milk	118	89.8	118	94.9	3.6	0.058	118	91.5		0.655
A3	Pregnant women should not be given raw milk	118	69.5	118	91.5	17.8	<0.001	118	93.2	19.6	<0.001
A4	Boiling milk is something that should always be done to ensure the milk is safe for people	118	87.3	118	99.2	12.3	<0.001	118	94.1	3.2	0.074
A5	Boiling of milk changes the taste of milk[2]	117	6.0	117	3.4	1.3	0.257	117	7.7	0.3	0.593
A6	Boiling of milk prevents *gullichoo* (milk curd in the stomach of young children)[2]	118	0.8	118	2.5	1.0	0.317	118	0	5.4	0.020
A7	If food is clean to the naked eye observation, it is fully safe for human consumption[2]	117	57.3	114	75.4	7.1	0.008	118	74.6	8.7	0.003
A8	Milk from animal suffering of mastitis is bad for human consumption	118	76.3	118	94.1	16.3	<0.001	118	89.8	8.5	0.004
A9	Milk of animal treated with "white" or "black" medicine antibiotic should not be used for human consumption[3]	118	55.9	118	84.7	23.1	<0.001	118	90.7	30.6	<0.001
A10	"Evil eye" (*buda*) causes mastitis in livestock[2]	118	11.0	116	42.2	32.4	<0.001	118	20.3	3.9	0.048

[1]Where n = number of respondents with valid responses to each question, and % = proportion of women responding correctly to the statement.
[2]Indicates negatively worded statements (i.e., disagreement is a positive attitude).
[3]"White" and "black" medicine, respectively, refer to injection of penicillin-streptomycin and oxytetracycline (2 antibiotics commonly used in the treatment of livestock in the area).

sessment and investigate whether the training led to changes in the responses to the statements. An overall knowledge score was calculated as the mean of the "assessment score" achieved by each woman. Specifically, for each woman, an assessment score (percentage) was calculated by dividing the number of correct answers provided by the total number of knowledge statements (8) and multiplied by 100 to express it as a percentage). The mean assessment score for the entire group was calculated pretraining, immediate post-training, and 6 mo post-training. The same was done to calculate overall attitude and practice scores. A paired *t*-test was used to compare the overall KAP scores between post-training assessments and pretraining; *P*-values < 0.05 were considered significant (95% confidence level).

RESULTS

Participants' Characteristics and Livestock Production Activities

Complete data were available for 118 trainees. We were unable to link the post-training form with the pretraining assessment of one participant (name of the respondent in the first and last evaluations was different) and were unable to interview one woman in the 6 mo post-training assessment as she was unavailable because of a medical condition. Out of the 118 women with complete records, 47.4% were under 30 yr of age, 37.3% were between 31 and 50 yr old, and 15.3% were older than 50 yr. Only 4.2% of the women could read and write, and 8.5% of them had previously attended training on milk hygiene and quality management or good hygienic dairy production. Cattle were the most common livestock species kept by the study participants' households (Table 4).

Effect of Training on the Knowledge of Pastoral Women Regarding Milk Hygiene and Zoonoses Prevention

Table 1 shows the change from pre- to post-training in the knowledge of women regarding food hygiene and the prevention of zoonoses. For most of the knowledge questions, there was a significant increase in the percentage of women able to provide a correct answer in the assessments after training compared with before training. However, there was no significant change in responses to 2 of the statements: (1) "Eating food contaminated with feces from animals can make people sick," and (2) "All people, including children, adults, pregnant women and elders, are at equal risk of foodborne disease." Regarding potential transmission of zoonotic diseases (statements K7 and K8, Table 1),

Table 3. Practices of pastoral women regarding milk hygiene and zoonotic diseases: comparing post-training evaluation with pretraining evaluation (percentage of women reporting adopting each practice)[1]

Code	Practice questions	Pretraining		6 mo post-training		McNemar	
		n	%	n	%	χ^2	*P*-value
P1	How often do you wash your hands before milking cows?	118	69.5	118	87.3	9.8	0.002
P2	How often do you wash your hands after milking cows?	116	81.0	118	91.5	5.8	0.002
P3	Do you discard the first squirts of milk when milking animals?	118	28.0	118	53.4	17.3	<0.001
P4	How often do you boil milk before consumption?	118	29.7	118	50.8	11.4	<0.001
P5	Do you prevent calf from suckling when milking cows or camels?	116	29.3	117	51.3	13.0	<0.001
P6	Do you prevent yourself or a family member from handling milk when the individual has diarrhea?	118	39.8	118	84.7	43.2	<0.001
P7	Do you prevent yourself or a family member from handling milk when the individual has a longstanding cough?	118	34.7	118	83.1	48.5	<0.001
P8	Do you prevent yourself or a family member from milking cows when the individual has a wound on his or her hand?	118	33.1	118	80.5	47.5	<0.001
P9	Do you avoid using milk of animals receiving antibiotic (black or white medicine[2]) injection?	117	53.0	114	93.9	39.7	<0.001

[1]Where n = number of respondents with valid responses to each question, and % = proportion of women reporting adopting each practice.

[2]"White" and "black" medicine, respectively, refer to injection of penicillin-streptomycin and oxytetracycline (2 antibiotics commonly used in the treatment of livestock in the area).

the participants had relatively low knowledge before training (54.2% gave correct answers to questions on tuberculosis, and 57.6% gave correct answers to questions on diseases transmitted to humans as a result of exposure to aborted materials likely due to brucellosis). There was a significant increase in participants' knowledge of animal-to-human transmission of zoonotic diseases. The evaluation of the overall knowledge score (Table 5), showed a significant ($P < 0.001$) increase in the knowledge of the participants, from an overall pretraining score of 75.6% to 91.4% at the immediate post-training assessment and to 90.0% at 6 mo post-training.

Effect of Training on the Attitudes of Pastoral Women Regarding Milk Hygiene and Zoonoses Prevention

Table 2 presents the results of attitude assessment before and after training. The statement with the highest proportion of women displaying a "positive attitude" was "Children should not be given raw milk" (statement A2, Table 2) and the one with lowest proportion of women with "positive attitude" was "Boiling of milk prevents *qullichoo* (milk curd in the stomach of young children)" (statement A5, Table 2). In the immediate post-training evaluation, the percentage of the trainees displaying positive attitudes to the statements was higher than that before training for all statements except one—the statement of the effect of boiling milk on its taste (statement A5, Table 2) showed a small decrease in the proportion of women displaying a positive attitude at the immediate post-training evaluation and a small increase in that proportion 6 mo post-training. There was a low level of positive attitude toward this statement (i.e., most women responded that boiling changes the taste of milk) and the training did not cause a statistically significant change at either time of evaluation. At baseline (pretraining), 76.3% of respondents had moderately positive attitudes regarding the unsuitability for human consumption of milk from lactating animals suffering from mastitis (statement A8, Table 2) and the proportion of respondents displaying a positive attitude increased after the training. The level of positive attitude related to consumption of milk of lactating animals being treated with antibiotics (statement A9, Table 2) was low in the pretraining assessment (55.9%) and increased significantly immediately after training (84.7%) and 6 mo later (90.7%). In contrast, only 11%

Table 4. Species of livestock kept by the households of the participants (n = 118)

Species	Households keeping (%)	Number per household		
		Mean	Minimum	Maximum
Cattle	96.6	11.2	0	100
Lactating cows	83.1	2.1	0	10
Goats	94.1	13.9	0	90
Lactating goats	88.2	5.1	0	30
Camels	41.5	1.5	0	10
Lactating camels	27.1	0.5	0	4
TLU[1]	100.0	11.4	0.6	86

[1]TLU = tropical livestock unit (equivalent to a hypothetical animal of 250 kg), conversion factors: 1 head of cattle = 0.7 TLU, 1 head of sheep/goat = 0.1 TLU, and 1 head of camel = 1 TLU (Jahnke, 1982).

Table 5. Mean percentage of correct answers for knowledge questions (n = 8), positive attitude (n = 10) and correct practices (n = 9) of the training participants

	Correctly answered statements (% mean ± SD[1])			Immediate post- vs. pretraining		6 mo post- vs. pretraining	
KAP	Pretraining	Immediate post-training	6 mo post-training	Paired *t*-test	*P*-value	Paired *t*-test	*P*-value
Knowledge	75.6 ± 1.7	91.4 ± 0.7	90.0 ± 0.7	9.4	<0.001	7.9	<0.001
Attitude	58.8 ± 1.6	64.7 ± 1.2	61.4 ± 0.9	3.2	0.001	1.4	0.08
Practice	49.5 ± 2.7	—	64.7 ± 1.8	—	—	5.4	<0.001

[1]Standard deviation of the percentage.

of participants disagreed with the statement "'evil eye' causes mastitis," which corresponds to a positive attitude (statement A10, Table 2) at baseline, and the percentage increased to 42.2% immediately after training ($P < 0.001$) and to 20.3% 6 mo after training ($P = 0.048$). The percentage of women displaying a positive attitude to the statement of not giving raw milk to children (statement A2, Table 2) was already high at baseline and this percentage increased only slightly at post-training evaluations. We found a statistically significant difference in the overall attitude score at the immediate post-training evaluation (64.7%) compared with the pretraining score (58.8%) but not 6 mo after the training (61.4%; Table 5).

Effect of Training on Self-Reported Practices

Table 3 shows the results of participants' self-reported practices before and 6 mo after the training. At pretraining, 30% of participants reported boiling milk before consumption, increasing to 51% 6 mo after training. For all practices, we observed a statistically significant increase in the percentage of women reporting undertaking such practice. The overall practice score increased from 49.5% (pretraining) to 64.7% (6 mo post-training) and the change was statistically significant (Table 5).

DISCUSSION

Hygienic handling of milk and milk products along the value chain is necessary to maintain product safety and quality and protect the health of consumers. This study evaluated the effectiveness of training in improving KAP of pastoral women with regards to hygienic handling and consumption of milk and milk products and preventing transmission of milk-borne zoonoses

We found that a face-to-face training that delivered locally tailored content improved the knowledge and attitudes of pastoral women around hygiene of milk and milk products during handling and processing. Immediately after the training, we observed an increase in the percentage of women responding correctly to the knowledge statements and an increase in the overall knowledge score; this effect was also present 6 mo after training, suggesting that knowledge can be sustained at least in the medium term. This is a promising finding, but follow-up studies are required to confirm the medium- to long-term impacts of training and what factors determine knowledge retention in the long run. Some studies in general food hygiene considered changes registered 3 to 5 mo after training as medium term and those registered up to 2 yr as long term (Touré et al., 2011; Takanashi et al., 2013).

The change in knowledge, however, was not observed for all statements included in the evaluation of the training. For example, the disproportionate susceptibility of different population groups to food-borne disease (i.e., the high susceptibility of pregnant women, young children, and the elderly to food-borne infection) was not recognized by participants in the pretraining assessment and there was no change in this knowledge after training. This may be due to the fact that some health problems lack a corresponding terminology in the Borana local language, which may complicate the transfer of knowledge of certain concepts during training. This has been reported for other traditional communities; a study conducted in Tanzania showed a lack of local concepts for most zoonotic diseases in the local language of a typical traditional pastoral society (Mangesho et al., 2017). Alongside that, in populations with limited formal education, it can be difficult to explain complex biological processes of health and disease; for example, the concepts of immunological status and disease susceptibility of different population groups. The health risk of animal fecal contamination of food was understood correctly by majority of the respondents even before the training was given. It can be difficult to explain the exact reason for this high rate of correct answers, whether it was related to the knowledge of the exact risk from fecal bacteria or whether it might be due to aesthetic reasons from visible fecal matter on food.

Before training, the average attitude score was 59%, and this increased immediately after training but decreased over 6 mo post-training. The percentage of women reporting a "positive attitude" of not giving raw milk to children and to pregnant women at both post-training evaluations was higher than at baseline. Consumption of raw milk is a well-known source of food-borne disease, especially for children and vulnerable individuals, and acquiring knowledge on milk and food-borne disease seemed to have translated into a change in attitude on the adequacy of raw milk in the diets of children and pregnant women. Similarly, more women reported "positive attitudes" regarding boiling of milk to make it safe during the post-training evaluations compared with baseline.

However, women displayed other contrasting attitudes toward raw milk, such as in response to the statements on reasons and consequences of boiling milk. Participants' attitudes in relation to reasons for boiling milk (e.g., preventing formation of milk curd in children and changes in taste) were what our study considered "negative," which did not change after training. The inability of training to address those attitudes may relate to the fact that there is a deeply rooted, but not evidence-based, belief in Borana, that dictates that milk should be boiled before feeding it to children to prevent the formation of milk curd in the stomach of young children, which is believed to happen when children consume raw milk (Amenu et al., 2019b). Moreover, the training failed to change the misperception of participants related to the "evil eye" as the cause for mastitis in dairy animals. An earlier study (Amenu et al., 2017) in the same area reported the misperception by the local people that "evil eye" causes acute udder swelling (most probably due to mastitis). Mastitis is a common problem in lactating cows, especially where farm management practices are suboptimal. It is caused by ubiquitous bacteria, some of which are important milk-borne pathogens in humans. The misperceptions around the causes of mastitis and its relation to human health and the difficulty in improving the participants' understanding of the subject calls for continuous awareness creation in the pastoral community.

Interestingly, training seemed to positively affect participants' adoption of most practices; the percentage of self-reported correct practices after the training intervention in all the questions showed statistically significant changes. This is a promising result although at 6 mo after training, the practice score was 65%, showing that about one-third of practices were not adopted by participants, from a baseline of 50%. This finding reveals that, although promoting better practices can be done through training, sustaining the adoption of such practices over time may require additional efforts or a modified approach to training, both to improve the practice score and to sustain adoption of practices over time. At the same time, the limitations of infrastructure, access to equipment, suboptimal water quality, and access to sanitation facilities in the study area can limit the ability of the communities to put knowledge into practice. The failure to translate knowledge or attitudes into practices and to maintain changes in practices is a common problem found in food hygiene training evaluations (Morse et al., 2019). For example, in a recent integrated review of food hygiene training interventions at the global level, it was found that translation of knowledge into attitudes or practices or translation of attitudes into practices rarely happened after training; a lack of translation of knowledge into practices was observed in 50% (n = 36) of the articles used in the review (Zanin et al., 2017). Complementing theoretical sessions with practical sessions and motivating trainees during trainings can help improve the ability of trainings to result in changes in practices (da Cunha et al., 2013). Moreover, it has been suggested that conducting the trainings at the workplace of the trainees can improve translation of knowledge into practice (McIntyre et al., 2013). Further, for the effect of trainings to be sustainable and long-term, an enabling environment should be created by designing applicable policies and improving overall infrastructure (Grace et al., 2019), which are often lacking in remote and rural settings. In addition, identifying and addressing the social and economic constraints as well as leveraging incentives for adopting risk-mitigating practices are essential aspects that should be built in to capacity development initiatives. Incentives may be addressed through support in providing materials for hygiene and finding a market niche for improved and safe products.

The study results are based on the participants' self-assessment of KAP; it is likely that responses of the pastoralists to some of the statements could be inaccurate (i.e., over- or understated) due to social norms or the pressure to answer "correctly." Using a checklist to record actual practices of participants, as opposed to relying on self-reports, is a more objective way of recording this information; however, in pastoralist contexts, this has its own challenges due to the scattered residence patterns, which make farm-level observations very difficult in terms of time and logistics. In addition to improving the accuracy of the KAP measures, future studies should consider assessing the impact of training on milk quality and safety through microbiological assessments of milk and milk products and observation of the hygienic status of farms, aspects that were not included in this study and on which evidence is scarce.

CONCLUSIONS

A training intervention for safe milk handling and processing significantly improved the KAP of participants for most of the questions and statements assessed. In particular, training was successful in improving the knowledge, attitude, and self-reported practices of pastoral women with regard to milk hygiene and milk-borne zoonoses. However, negative attitudes and incorrect practices in relation to milk safety remained in a large proportion of participants. Although training can be used as a vehicle to improve KAP, the design of the trainings needs to consider innovative approaches that would promote the adoption of good hygiene practices and ensure the sustainability of such adoption over time. This could include practical sessions in the homestead or herding places of the pastoralists in collaboration with the local government. In general, integrated approaches are in development for sustainable improvement of the hygiene and safety of milk and milk products. To bring more efficient and sustainable impact to the food safety practices in traditional milk value chains, training interventions need to be supported by the introduction of locally adaptable technologies, by providing incentives and by creating an enabling environment, such as overall infrastructure development, including improving water quality and sanitation facilities.

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